

#define [citation needed?] If elaboration needed make a note otherwise you can ignore :D

3.5 hours on examples

Problem 1 (95SLS3)

For each non-negative n , we let $\{a_0, a_1, \dots\}$ be its digits reversed when written in binary with infinitely many leading zeroes. In other words,

$$n = a_0 2^0 + a_1 2^1 + a_2 2^2 + \dots$$

We claim that, where $\{a_0, a_1, \dots\}$ is such a sequence for n , we have,

$$x_n = p_0^{a_0} \cdot p_1^{a_1} \cdot p_2^{a_2} \cdot \dots$$

We prove this using induction... Very standard induction to finish.

Problem 2 (IMO 2017/1)

We claim that a_0 is sufficiently and necessarily be a multiple of 3.

Lemma: Where $f(n)$ gives the smallest $m = n + 3k$ that is a square for some non-negative integer k , we have $\sqrt{f(n)} < n$ for all $n > 3$ and $n \not\equiv 2 \pmod{3}$.

Proof:

If $n \equiv 0 \pmod{3}$, then the square root of $f(n)$ is the smallest multiple of three greater than $\lceil \sqrt{n} \rceil$ [citation needed?]. For example, for $n = 6$, $\lceil \sqrt{n} \rceil = 3$. Therefore, the square root of $f(n) = 9 < 6$. For $n \geq 9$, this quantity is less than $\lceil \sqrt{n} \rceil + 3 < \frac{n}{2} + 3 < n$ obviously.

If $n \equiv 1 \pmod{3}$, then the square root of $f(n)$ is the smallest non-multiple of three greater than $\lceil \sqrt{n} \rceil$. We may test $n = 4, 7$ to see $\sqrt{f(n)} = 2, 4$ respectively, which both works. For $n > 9$, this quantity is once again less than $\lceil \sqrt{n} \rceil + 3 < \frac{n}{2} + 3 < n$ obviously.

Claim: When we have $a_k \equiv 2 \pmod{3}$ for some k , it does not work.

Proof: Squares cannot be $2 \pmod{3}$ so we'll be stuck in $2 \pmod{3}$ with a monotonic increasing sequence.

With the lemma proven, we use induction.

For $a_0 \equiv 0 \pmod{3}$, we claim it always works.

Base case: $a_0 = 3 \Rightarrow a_1 = 6 \Rightarrow a_2 = 9 \Rightarrow a_3 = 3 \dots$ We have a repeat.

Inductive step: For $a_0 > 3$, we follow the process until we get to $\sqrt{f(a_0)} < a_0$, then use induction hypothesis.

Conclusion: By induction we're done.

For $a_0 \equiv 1 \pmod{3}$, we claim it never works.

Base case: $a_0 = 1 \Rightarrow a_1 = 4 \Rightarrow a_2 = 2$. Using the claim, we're done.

Inductive step: For $a_0 > 3$, we follow the process until we get to $\sqrt{f(a_0)} < a_0$, then use induction hypothesis.

Conclusion: By induction we're done.

6 clubs; 4.5 hours

Problem 8 (19IMO5)

We claim the expected number of moves is $\frac{n(n+1)}{4}$. Proving this immediately gives the number of steps for every start is finite as desired. We prove this with induction.

Base case: $n = 1$

Obviously, H needs one step and T needs none for expected value $\frac{1}{2} = \frac{1(1+1)}{4}$ as desired.

Inductive step: $n = k \rightarrow n = k + 1$

We may consider the directed graph G_k where the 2^k nodes are all the possible ways the coins may be flipped. We then let there be a directed edge from each node (except for $TT \dots T$) to its state after Harry does one operation.

We label each edge by the number of directed edges we must pass through starting from there before getting to $TT \dots T$ walking on G_k . By the induction hypothesis, the average of these labels is $\frac{k(k+1)}{4}$.

We construct G_{k+1} .

For the 2^k of the 2^{k+1} nodes that end with T , we claim that they just form G_k except every node has an extra T at the end. For every such node on G_{k+1} , the coin to flip is still the same as there's the same amount of H 's on the board. The final T would never be flipped since that'd require $k + 1$ H 's on the board, so it's just the same graph except with an extra T at the end.

The labels of these vertices are the same as before.

For the 2^k nodes that end with an H , we claim we can construct it by starting with G_k , flipping every coin, reversing the order of the coins, then adding an H at the end. Let n be the amount of H 's there originally was, we now have $k + 1 - n$ H 's. Therefore, if we were originally able to flip something, we'd flip the corresponding spot after having previously reversed the order of coins. Therefore, the result is still just the previous result after flipping every coin, reversing the order of coins, then adding an H at the end.

The labels of these vertices $k + 1$ more than before since we'd arrive at $HH \dots H$ after moving the original label number of moves, then require $k + 1$ moves to be done.

Therefore, the new expected value is,
$$\frac{\frac{k(k+1)}{4}2^k + \left(\frac{k(k+1)}{4} + (k+1)\right)2^k}{2^{k+1}} = \frac{(k+1)(k+2)}{4}.$$

Problem 9 (07SLC1)

We claim that there is only one solution. The construction is as follows: we first divide the sequence into $n + 1$ blocks on n consecutive terms starting with $a_1, \dots, a_n$. For the k -th block (zero-indexed), we have,

$$a_{nk+1} = a_{nk+2} = \dots = a_{nk+n-k} = 0; a_{nk+n-k+1} = a_{nk+n-k+2} = \dots = a_{nk+n} = 1$$

For example, for $n = 4$, we have,

$$\text{Block 0: } (a_1, \dots, a_4) = (0, 0, 0, 0);$$

$$\text{Block 1: } (a_5, \dots, a_8) = (0, 0, 0, 1);$$

$$\text{Block 2: } (a_9, \dots, a_{12}) = (0, 0, 1, 1)$$

$$\text{Block 3: } (a_{13}, \dots, a_{16}) = (0, 1, 1, 1)$$

$$\text{Block 4: } (a_{17}, \dots, a_{20}) = (1, 1, 1, 1)$$

Clearly, if we choose $i = xn + y$ for some non-negative integers x, y , we clearly have,

$$a_{i+1} + \dots + a_{(x+1)n} \leq a_{i+n+1} + \dots + a_{(x+2)n}$$

With equality case if and only if everything is one.

$$a_{(x+1)n+1} + \dots + a_{i+n} \leq a_{(x+2)n+1} + \dots + a_{i+2n}$$

With equality case if and only if everything is zero.

However, these cannot both take the equality case since different rows [citation needed? These 5 lines]. Adding these, we are done.

We now prove uniqueness. First, since,

$$a_1 + a_2 + \dots + a_n < a_{n+1} + \dots + a_{2n} < \dots < a_{n^2+1} + a_{n^2+2} + \dots + a_{n^2+n}$$

Each of these expressions must be an integer on $[0, n]$. Therefore, by Pigeonhole,

$$a_1 + a_2 + \dots + a_n = 0;$$

$$a_{n+1} + \dots + a_{2n} = 1;$$

...

$$a_{n^2+1} + \dots + a_{n^2+n} = n$$

In fact, these are the $n + 1$ blocks.

We use induction on the blocks to prove that the construction presented is necessary.

Base Case: Block 0, since $a_1 + a_2 + \dots + a_n = 0$ we have they're all zero as desired.

Inductive Step (Block k to Block $k + 1$):

We have the binary string for block k looks like,

$$00 \dots 0 \underbrace{11 \dots 1}_k$$

We do yet another induction in block $k + 1$ to prove $a_{n(k+1)+m} = 0$ for $1 \leq m \leq n - (k + 1)$.

Let m be the minimum positive integer such that $a_{n(k+1)+m} = 1$.

FTSOC, let $m \leq n - (k + 1)$.

Claim: $a_{n(k+2)+m'} = 1$ for some positive $m' < m$.

We know,

$$a_{nk+m+1} + \dots + a_{n(k+1)+m} < a_{n(k+1)+m+1} + \dots + a_{n(k+2)+m}$$

By the induction hypothesis $a_{nk+m+1} + \dots + a_{n(k+1)} = k$, so the LHS is $k + 1$.

$$\Leftrightarrow k + 1 < a_{n(k+1)+m+1} + \dots + a_{n(k+2)+m}$$

$$\Leftrightarrow a_{n(k+1)+m+1} + \dots + a_{n(k+2)+m} \geq k + 2$$

However,

$$a_{n(k+1)+1} + \dots + a_{n(k+1)+n} = k + 1$$

$$\Leftrightarrow a_{n(k+1)+m+1} + \dots + a_{n(k+1)+n} = k$$

Therefore,

$$a_{n(k+2)+1} + \dots + a_{n(k+2)+m} \geq 2$$

If the claim isn't true, then the LHS is capped at one so impossible.

With the claim proven, we may recursively use it until we have, $a_{n^2+m'} = 1$ for some positive $m' < 1$.

[citation needed?] (!)

Therefore, with this contradiction, we have,

$$a_{n(k+1)+1} = a_{n(k+1)+2} = \dots = a_{n(k+1)+n-(k+1)} = 0$$

Since this block sums to $k + 1$, everything else must be one as desired.

Conclusion: By mathematical induction we are done.

16 clubs; 7.5 hours

Problem 12 (98SLN8)

We construct the sequence a , making necessary additions at each step. First, since 0 may be constructed, there must be a zero in a , meaning $a_0 = 0$.

Claim: If the smallest number the terms $a_0, \dots, a_n$ cannot express is k , then $a_{n+1} = k$.

If we fix $a_{n+1} < k$, then note that we already had a way of generating a_{n+1} . Additionally, we may now take $i = n + 1; j = k = 0$ for a new way, so this is not allowed.

If we take $a_{n+1} > k$, then obviously we won't be able to express k .

For example, using this claim, we'd get, $a_1 = 1$ as we still cannot generate anything but zero. With these two, we may then generate 0 ... 7 but not 8, so $a_2 = 8$.

We claim that, using this process, we recursively generate,

$$\begin{aligned} a_0 &= 0; \\ (a_0, a_1) &= (0, 1); \\ (a_0, a_1, a_2, a_3) &= (0, 1, 8, 9); \\ (a_0, a_1, a_2, a_3, a_4, a_5, a_6, a_7) &= (0, 1, 8, 9, 64, 65, 72, 73); \\ &\dots \end{aligned}$$

More specifically, we claim that $(a_{2^k}, \dots, a_{2^{k+1}-1}) = (a_0 + 8^k, a_1 + 8^k, \dots, a_{2^{k+1}-1} + 8^k)$ for $k \geq 0$ and $a_0 = 0$. We also claim that, with this, $(a_0, \dots, a_{2^k-1})$ generates all nonnegative integers strictly less than 8^k .

We've already proved $a_0 = 0$ and we will prove the rest using induction.

Base case: $k = 0$. We've already proved $a_1 = 1$ and that the first two generates everything strictly less than $8^1 = 8$.

Inductive step: $k \rightarrow k + 1$

We already have $(a_0, a_1, \dots, a_{2^k-1})$ generates all nonnegative integers strictly less than 8^k . Therefore, $a_{2^k} = 8^k = a_0 + 8^k$ as desired. *Sketch: For the rest, it's the same thing as before except we may choose how many 8^k s to add.*

Conclusion: By mathematical induction we're done.

We now extract a_{1998} . We claim that $a_n = b_0 8^0 + b_1 8^1 + b_2 8^2 + \dots$ where b is the binary representation of $n := b_0 2^0 + b_1 2^1 + \dots$. We may prove this by induction. [citation needed?]

Therefore, since $1998 = 2^{10} + 2^9 + 2^8 + 2^7 + 2^6 + 2^3 + 2^2 + 2^1$, the answer is,

$$8^{10} + 8^9 + 8^8 + 8^7 + 8^6 + 8^3 + 8^2 + 8^1$$

Problem 14 (APMO 2018/4)

(The entire proof has parts throughout that I feel is unsound but difficult to make rigorous further.)

We draw the path of the ray in red up to the point at which it reflects off a side, by which time the color changes to blue. If we flip the triangle over the side, it reflects off, the red ray in the original triangle will be on the same line as the blue line in the reflected triangle.

We create an infinite plane tiled by equilateral triangles. We label three vertices on the same equilateral triangle A, B, C . We then label the rest of the vertices such that all the other equilateral triangles also have vertices labeled A, B, C ; by induction there is a unique way to do this.

Any line segment drawn between two vertices labeled A on this plane that does not overlap with any side nor pass through any vertices will correspond to a ray shot out that returns to A . [citation needed?]

The vertices forms rows like so.

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BCABCABCABCABCABC
CABCABCABCABCABCAB
BCABCABCABCABCABC
CABCABCABCABCABCAB
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Notice if we dilate this vertically or horizontally, it wouldn't affect the result as it won't affect whether segments will pass through other vertices. We stretch it horizontally such that all original equilateral triangles are now isosceles right triangles.

We now put this on the coordinate plane. However, we impose the unit vectors such that $(0, 0)$ has label A , $(1, 0)$ has label B , and $(0, 1)$ has label C .

WLOG, we consider segments from $(0, 0)$ to (x, y) for some non-negative x, y .

By inducting on rows, coordinates (x, y) is labeled A if and only if $x \equiv y \pmod 3$. Additionally, a segment from $(0, 0)$ to (x, y) passes through no other vertices if and only if $\gcd(x, y) = 1$. Finally, traveling from $(0, 0)$ to (x, y) , we pass through $(y - 1)$ edges parallel to $\hat{i}$, $(x - 1)$ edges parallel to $\hat{j}$, and $(x + y - 1)$ other edges not counted for a total of $n = 2x + 2y - 3$.

We claim that $n \equiv 1, 5 \pmod 6$; $n \neq 5, 17$ work, and no other n work.

For the any other n , note that obviously $n = 2x + 2y - 3$ is odd. Additionally, if $n \equiv 0 \pmod 3$, then $x + y \equiv 0 \pmod 3$; $x \equiv y \pmod 3 \Rightarrow 3 \mid x, y$, which violates $\gcd(x, y) = 1$.

For $n = 5$, we'd need $2x + 2y = 8 \Leftrightarrow x + y = 4$. All five cases clearly fail at least one criterion.

For $n = 17$, we'd need $2x + 2y = 20 \Leftrightarrow x + y = 10$. Since $x \equiv y \pmod 3$, we may have $(2, 8)$; $(5, 5)$; $(8, 2)$ which all fail $\gcd(x, y) = 1$.

Now for the construction, if $n \equiv 1 \pmod 6$, we may simply take $x = 1$; $y \equiv 1 \pmod 6$.

For $n \equiv 11 \pmod 12$, we take $x = 2$; $y \equiv 5 \pmod 6$, which satisfies $\gcd(x, y) = 1$; $n \equiv 11 \pmod 12$; $x \equiv y \pmod 3$.

For $n = 24k + 5$, we take $(6k + 5, 6k - 1)$.

For $n = 24k + 17$, we take $(6k + 11, 6k - 1)$.

Problem 15 (15EGMO2)

We divide the $2n \times 2n$ chessboard into n^2 disjoint 2×2 squares, which we call the best squares. By Pigeonhole, since we have $2n^2$ cells covered in total, and each 2×2 square has at least 2 unit squares

covered, each must have exactly two. In addition, the squares not covered must be in the same row/column.

Claim: We cannot have any dominoes partially covering two best squares.

FTSOC, say that we do. WLOG, let's say a domino is placed horizontally partially covering the top row of two best squares.

	*	*	

For the right best square being covered, we need to cover another cell adjacent to the first. If we place another domino vertically that covers the square, we'd have a 2×2 square that contains only 1 uncovered unit square (!). Therefore, we must place yet another domino horizontally directly to the right of the first. However, inducting, we'd eventually reach the edge when this becomes impossible.

Claim: There's a bijection between selecting which squares to cover and valid domino placement.

We look at which two are covered in each best square. Using the previous claim, we must place a domino there.

Normal finish

Checkboard color the board, the way to color to black squares is independent from that of the white squares. By 19CAN3, each are $\binom{2n}{n}$ so answer is $\binom{2n}{n}^2$.

Alternate finish?

We label each of the best squares L (left), U (up), R (right), D (down) for which edge both covered cells of the best square both the covered cells are adjacent to.

We check the conditions for each 2×2 square satisfying at least two uncovered unit squares which lie in the same row or column. For the 2×2 squares that cover two best squares, they work if and only if,

For all horizontally adjacent, we only have LR, UR, DR, LU, LD, LL, RR, DD, UU.

For all vertically adjacent, we only have $\begin{matrix} U & U & U & L & R & L & R & D & U \\ D' & L' & R' & D' & D' & L' & R' & D' & U' \end{matrix}$.

Following these rules, for the 2×2 squares that cover four best squares, we may casework to show that satisfying the previous conditions is sufficient.

Case 1: More than three are covered.

WLOG, let the top-left, top-right, and bottom-left be covered. Clearly, we cannot cover the one above the top-left for there'd be another 2×2 square with three covered. Therefore, the one to the left of the top-left must be covered. However, in that case, the 2×2 square one unit to the left has three covered (!).

Case 2: Two are covered in opposite corners.

WLOG, let the bottom-left and top-right be covered. Considering the four best squares being partially covered, the top two may form LD, LL, DD. The bottom two may form RR, UR, UU. Every one of the 9 combinations fails the vertically adjacent rule.

We now count how many ways there are to fill a $n \times n$ grid with L, R, U, D following these rules... **Not sure if there's a way to finish this line of thought?**

29 clubs; 12 hours

Problem 16 (15AMO3)

Case 1: The empty set is blue.

Let K be a set of integers on $[1, n]$ such that an integer $x \in K$ if and only if $\{x\}$ is colored blue on S . We claim that, in fact, for any K , a set $T \subset S$ is colored blue if and only if T is a subset of K . Therefore, each of the 2^n ways to color the single element sets determines the entire coloring.

We prove the if part by induction on the cardinality of each subset.

Base case: Subsets of size one. This is already given.

Inductive Step: Subsets of size $k \rightarrow k + 1$.

For the subsets of size $k + 1$ that are subsets of K , we may find two subsets T_1, T_2 that satisfy $|T_1| = |T_2| = k$ and $T_1 \cup T_2$ is the subset in question.

$$f(T_1)f(T_2) = f(T_1 \cup T_2)f(T_1 \cap T_2)$$

Obviously, T_1, T_2 are all subsets of K .

$$\Leftrightarrow 2^k 2^k = f(T_1 \cup T_2) 2^{k-1}$$

$$\Leftrightarrow f(T_1 \cup T_2) = 2^{k+1}$$

Since there were this many subsets of $T_1 \cup T_2$ in the first place, every one of them must be colored blue.

Conclusion: By induction we're done.

To prove only if, if a set T is not a subset of S , we start with having all subsets of $T \cap S$ being blue. Then, we plug in $T_1 = T \cap S; T_2 = \{x\}$ for some $x \in T; x \notin S$. We have,

$$f(T_1)f(T_2) = f(T_1 \cup T_2)f(T_1 \cap T_2)$$

$$\Leftrightarrow 2^{|T \cap S|} \cdot 1 = f(T_1 \cup T_2) \cdot 1$$

Therefore, as many subsets of $T_1 \cup T_2$ is colored blue as T_1 . Inducting, as many subsets of T is colored blue as that of $T \cap S$, meaning T cannot be colored blue.

Once again, in total we have 2^n cases here.

Case 2: The empty set is red. There is at least one set colored blue.

We claim that, in each coloring, if X is the smallest subset colored blue and Y is the largest, then a set is colored blue if and only if it is a subset of Y and a superset of X .

We first check that all such sets are indeed blue. Note that we may effectively erase X from every subset of S . Then, the empty set is blue and $Y - X$ is blue. Using the previous case as a lemma, this is possible if and only if the set containing a single element for every element in $Y - X$ is colored blue. In this case, all subsets of $Y - X$ are colored blue. [citation probably needed?]

Now we prove all other T are red.

Note that for any $|T| \leq |X|$. We defined X as the smallest blue set. If $|T| = |X|$ but $T \neq X$, then $f(T \cap X) > 0$. (!)

We look at the case where T is not a superset of X .

$$f(T) \cdot f(X) = f(T \cap X) \cdot f(T \cup X)$$

Since $|T \cap X| < |X|$, we have $f(T \cap X) = 0$. However, $f(X) = 1$ so $f(T) = 0$. Therefore, T must be colored red.

The case where T is not a subset of Y is analogous from the other direction.

Therefore, we may fix a choice of a pair $X \subset Y$ subsets of $\{1, 2, \dots, n\}$ with $|X| > 0$. To choose this, for each element, we may choose whether it is in Y , X (and Y), or neither X nor Y . That'd be 3^n possibilities, but there must be at least one element that's neither in X nor Y for $|X| > 0$ so in actuality it is $3^n - 2^n$.

Case 3: Everything is red.

One case.

In total, we have $3^n + 1$ cases.

38 clubs; 14 hours